COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	**CPE 010 - DATA STRUCTURES AND ALGORITHMS**	**SECTION:**	CPE21S4
DATE PERFORMED:	9 / 4 / 26	**DATE SUBMITTED:**	9 / 4 / 26
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Activity NO. 7.1 Sorting Algorithms

1. PROCEDURE OUTPUT

What procedures did you perform?

Code + Console Screenshot	
Observation	The array populated a 100 integer array with random values from 0 to 99 using rand() % 100. The output displays an unsorted dataset where elements appear in order.

```
int main() {
    int unsortedArr[maxSize];

    srand(time(0));

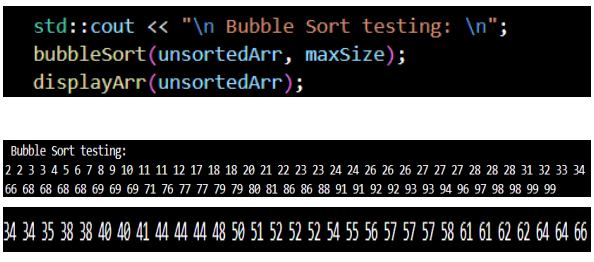
    for (int i = 0; i < maxSize; i++) {
        unsortedArr[i] = rand() % 100;
    }

    displayArr(unsortedArr);
```

```
19 80 0 28 38 11 81 65 99 30 68 90 26 99 52 39 68 68 35 8 6 44 86 7 76 74 74 79 46 2 41 57 71 88
59 26 7 18 63 5 68 79 66 24 67 40 63 71 31 56 86 26 70 75 85 19 66 55 12 41 37 8 99 95 59 17 85
PS C:\Users\Rhianne\Documents\DSA\DSA>
```

```
5 57 55 29 11 27 36 15 82 14 29 92 84 41 11 26 81 56 21 77 92 54 38 94 87 68 8 27 86 20 23 15 90
```

Table 7-1. Array of Values of Sort Algorithm Testing

Code + Console Screenshot	

```
std::cout << "\n Bubble Sort testing: \n";
bubbleSort(unsortedArr, maxSize);
displayArr(unsortedArr);
```

```
Bubble Sort testing:
2 2 3 3 4 5 6 7 8 9 10 11 11 12 17 18 18 20 21 22 23 23 24 24 26 26 26 27 27 27 28 28 28 31 32 33 34
66 68 68 68 68 69 69 69 71 76 77 77 79 79 80 81 86 86 88 91 91 92 92 93 93 94 96 97 98 98 99 99
```

```
34 34 35 38 38 40 40 41 44 44 44 48 50 51 52 52 52 54 55 56 57 57 57 58 61 61 62 62 64 64 66
```

Observation	Bubble Sort successfully sorted the array in ascending order by repeatedly comparing adjacent elements and swapping them when out of order, effectively pushing larger values to the right side of the list.

Table 7-2. Bubble Sort Technique

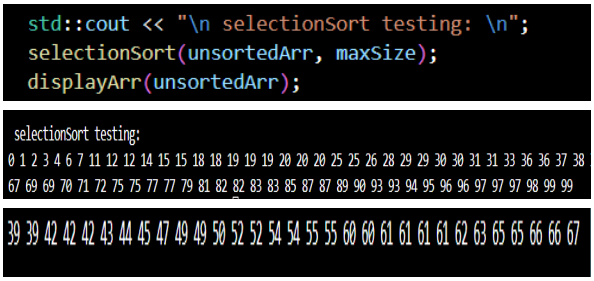
Code + Console Screenshot	
Observation	Selection Sort ordered the array by using routineSmallest() to find the minimum value in the unsorted section per pass, swapping it into its correct left-to-right position.

Table 7-3. Selection Sort Algorithm

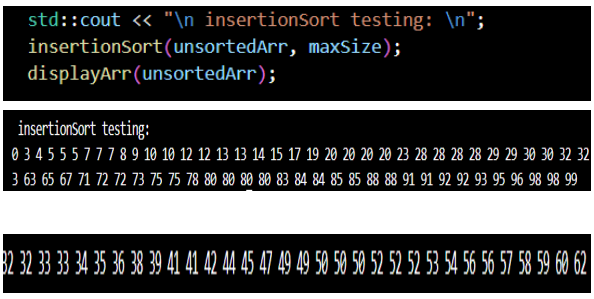
Code + Console Screenshot	
Observation	Insertion Sort built the sorted array element-by-element by holding the current key in temp and shifting larger preceding values to the right before inserting the key into its proper position.

Table 7-4. Insertion Sort Algorithm

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

CODE

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
#include <string>
#include "sortingAlgo.h"

const int VOTE_COUNT = 100;

struct Candidate {
    int id;
    std::string name;
    int votes;
};

int main() {
    Candidate candidates[5] = {
        {1, "Bo Dalton Capistrano", 0},
        {2, "Cornelius Raymon Agustin", 0},
        {3, "Deja Jayla Banaga", 0},
        {4, "Laila Brielle Yabut", 0},
        {5, "Franklin Relano Castro", 0}
    };

    int votesArr[VOTE_COUNT];

    srand(time(0));

    for (int i = 0; i < VOTE_COUNT; i++) {
        votesArr[i] = (rand() % 5) + 1;
    }

    std::cout << "Unsorted Votes:\n";

    for (int i = 0; i < VOTE_COUNT; i++) {
        std::cout << votesArr[i] << " ";
    }
```

```
38      std::cout << "\n\n";
39
40      insertionSort(votesArr, VOTE_COUNT);
41
42      std::cout << "Sorted Votes:\n";
43
44      for (int i = 0; i < VOTE_COUNT; i++) {
45          std::cout << votesArr[i] << " ";
46      }
47
48      std::cout << "\n\n";
49
50      for (int i = 0; i < VOTE_COUNT; i++) {
51          if (votesArr[i] == 1) {
52              candidates[0].votes++;
53          }
54          else if (votesArr[i] == 2) {
55              candidates[1].votes++;
56          }
57          else if (votesArr[i] == 3) {
58              candidates[2].votes++;
59          }
60          else if (votesArr[i] == 4) {
61              candidates[3].votes++;
62          }
63          else if (votesArr[i] == 5) {
64              candidates[4].votes++;
65          }
66      }
67
68      std::cout << "Vote Count:\n";
69
70      for (int i = 0; i < 5; i++) {
71          std::cout << "Candidate " << candidates[i].id
72                    << " - " << candidates[i].name
73                    << ": " << candidates[i].votes
```

```
74                 << " votes\n";
75         }
76
77         int winner = 0;
78
79         for (int i = 1; i < 5; i++) {
80             if (candidates[i].votes > candidates[winner].votes) {
81                 winner = i;
82             }
83         }
84
85         std::cout << "\nWinner: Candidate "
86                   << candidates[winner].id
87                   << " - " << candidates[winner].name
88                   << " with " << candidates[winner].votes
89                   << " votes\n";
90
91         return 0;
92     }
```

```
1  #ifndef SORTINGALGO2_H
2  #define SORTINGALGO2_H
3
4  template <typename T>
5  void insertionSort(T arr[], int maxSize) {
6      for (int k = 1; k < maxSize; k++) {
7          T temp = arr[k];
8          int j = k - 1;
9
10         while (j >= 0 && temp < arr[j]) {
11             arr[j + 1] = arr[j];
12             j--;
13         }
14
15         arr[j + 1] = temp;
16     }
17 }
18
19 #endif
```

OUTPUT

```
Unsorted Votes:
5 4 2 2 2 5 5 5 3 4 4 3 1 4 3 5 2 3 1 3 4 2 2 2 3 2 3 5 5 2 5 1 1 5 1 4 4 1 3 2 4 1 1
 3 4 5 4 3 1 2 3 2 4 5 4 3 1

Sorted Votes:
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3
 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

Vote Count:
Candidate 1 - Bo Dalton Capistrano: 15 votes
Candidate 2 - Cornelius Raymon Agustin: 20 votes
Candidate 3 - Deja Jayla Banaga: 23 votes
Candidate 4 - Laila Brielle Yabut: 24 votes
Candidate 5 - Franklin Relano Castro: 18 votes

Winner: Candidate 4 - Laila Brielle Yabut with 24 votes
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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